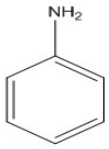
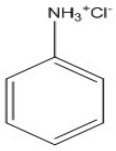
St. Philomena's College (Autonomous), Mysore
PG Department of Chemistry
Question Bank (Revised Curriculum 2018 onwards)
Second Year- Third Semester (2019 -21 Batch)
Course Title (Paper Title):
ORGANIC SPECTROSCOPY QP Code:55202

Unit	SI. No	Questions	Marks
1	1.	Calculate the wave number of radiation whose wavelength is 4 μm.	2
1	2.	Calculate the frequency of radiation whose wavelength is 400 nm. Express this wavelength in wave number	2
1	3.	Calculate the frequency of the radiation in hertz of wavelength 350 nm	2
1	4.	Calculate the wave number of the radiation if the frequency is 2.06×10^{14} Hz	2
1	5.	Calculate the wavelength in 'nm' if the frequency of strong yellow line in the spectrum is 5.09×10^{4} s^{-1}	2
1	6.	Determine the wavelength of electron moving with one third the speed of light	2
1	7.	Calculate the wavelength, wavenumber and frequency of radiation having an energy of 1.5 eV	2
1	8.	For infrared radiation of 5.00 μm, what is the wave number in cm^{-1}	2
1	9.	Calculate the quantum of energy, E in joules, J mol^{-1}, cal mol^{-1} and eV for a wavelength of 200 nm	2
1	10.	What is meant by polarised light?	2
1	11.	What is meant by specific rotation and molecular rotation?	2
1	12.	What are optically active molecules? Explain the condition for a compound to be optically active	2
1	13.	Define the term optical rotatory dispersion.	2
1	14.	Define circular dichroism? How is different from optical rotatory dispersion	2
1	15.	What is cotton effect?	2

1	16.	Explain octant rule. Give an example of this rule.	2
1	17.	What are the applications of optical rotatory dispersion and circular dichroism spectroscopy	2
1	18.	Why is ethanol is a good solvent in ultraviolet	2
1	19.	What are auxochromes and chromophores? Give an example for each.	2
1	20.	Explain why it is necessary to use quartz cuvettes while recording UV spectra.	2
1	21.	Which of the following compounds have higher absorption maximum	2

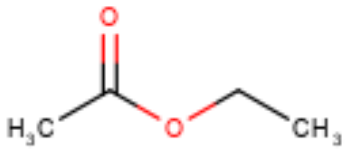
1 

2 

2	22.	How many normal vibration modes are possible in the linear molecule ethyne and non-linear molecule benzene	2
2	23.	Predict the number and give the names of the fundamental modes of vibration of HCl	2
2	24.	How many fundamental vibrational frequencies would you expect to observe in the IR absorption spectrum of water?	2
2	25.	How may fundamental vibrational frequencies would you expect to observe in the IR spectrum of carbondioxide	2
2	26.	How many fundamental modes of vibration would you predict for a)benzene b) toluene	2
2	27.	Why water can’t be used as a solvent for IR spectroscopy	2
2	28.	Which crystals are used for optical systems in IR spectroscopy? Justify your answer with a suitable explanation.	2
2	29.	Which selection rule is used in IR spectroscopy? Explain with suitable explanation supporting you answer	2
2	30.	What is the requirement for a vibration to be Raman active	2
2	31.	Why are the homonuclear diatomic molecules IR inactive and Raman active	2

2	32.	Explain why in Raman spectra the Stoke's lines are far more intense than the anti-stoke's lines, which are sometimes too weak for observation	2
2	33.	What are the differences between Raman spectra and IR spectra.	2
2	34.	Define rule of mutual exclusion principle with an example.	2
2	35.	State mutual exclusion principle for deciding IR and Raman activities.	2
2	36.	What is meant by functional group region of an IR spectrum? How is it useful in deciphering the structure of an organic molecule?	2
2	37.	In cyclic ketones, C=O absorption increases as the size of the ring decreases. Why?	2
2	38.	State ketones having minimum value of C=O absorption	2
2	39.	In halogenated ketones C=O absorption is 20 cm^{-1} higher than the corresponding non-halogenated compounds. Why?	2
2	40.	Distinguish between the following two isomers of compound namely propanal and acetone using IR spectroscopy	2
2	41.	Determine the structure of the compound, C_3H_6O showing absorption a) IR: 2720 and 1738 cm-1 b) UV : λ_{max} 292 nm, εmax 21	2
4	42.	How many signals does the molecule $(CH_3)_3CCH_2CHO$ have in Proton and ^{13}CNMR Spectra?	2
3	43.	What is meant by (n+1) rule in spin-spin coupling?	2
3	44.	Predict ^{1}H NMR splitting pattern for ortho- and para-methoxybenzaldehyde.	2
3	45.	Sketch the Proton NMR spectrum of ethyl iodide.	2
3	46.	Write the proton NMR Signal for the following compounds:	2
3	47.	Why TMS is used as standard in NMR?	2
3	48.	Why ethanol and water are not suitable solvents for recording ^{1}H NMR of organic compounds.	2
4	49.	Deduce the structure of organic compound ($C_{10}H_{14}$) with a triplet at δ 1.25 (6H), quartet 2.60 (4H), singlet 7.05(4H).	2
3	50.	A Peak appears at 240 Hz away from TMS in 60 MHz spectrometer. Find	2

		the chemical shift values in ppm.	
3	51.	Give the probable chemical shift and splitting pattern for CH_3-CH_2-CO-CH_3	2
4	52.	A compound $C_{10}H_{13}Cl$ shows following signals: 1.57 (s, 6H), 3.07(s, 2H), 7.27(S, 5H); Deduce the structure of compound.	2
4	53.	Give reason why a base peak is observed at m/z 120 in the mass spectrum of O-nitrotoulene which is not observed in case of p-nitrotoulene.	2
4	54.	Sketch ^{13}C resonance in pyridine and identify the C atoms showing the higest and lowest δ value.	2
4	55.	Why M^+ and (M^++2) peaks for CH_3Br will be almost in 1:1 ratio?	2
4	56.	Mention the main applications of ^{31}P NMR Spectra.	2
4	57.	Explain why ^{13}C shows NMR signals but ^{12}C does not?	2
4	58.	Name the reference compounds used in ^{31}P and ^{19}F NMR.	2
4	59.	What are metastable ions? Mention their importance.	2
4	60.	Write a note on ^{19}F NMR spectra for ClF_3	2
4	61.	Write a note on ^{19}F NMR spectra for SF_4	2
4	62.	Mention the applications of MRI	2
4	63.	Which reference compounds are used in ^{15}N and ^{19}F NMR spectroscopy	2
4	64.	State Nitrogen Rule with an example.	2
4	65.	Why M^+ and M^++1 peaks for CH_3Cl will be almost 3:1 ratio?	2
3	66.	Calculate magnetic field , B_0 corresponding to a precession frequency of 600MHz for 13C (γ= 267.53T-1. s-1)	2
3	67.	How many spin states possible for ^{1} H, ^{13}C, ^{35}Cl and ^{19}F.	2
3	68.	Calculate the chemical shift in ppm (δ) for a proton that has resonance at 240 Hz downfield TMS on spectrometer that operates at 200 MHz.	2
3	69.	How will you distinguish 1-Bromopropane and 2-Bromopropane using PMR spectroscopy?	2

3	70.	Define different type of relaxation processes during NMR Phenomenon.	2
3	71.	Why aromatic protons appear at higher δ value?	2
3	72.	Give reason for no signal for deuterium is observed in the PMR spectrum of a $CD_3CH_2CH_3$	2
3	73.	Differentiate chemically equivalent and magnetically equivalent protons taking suitable examples.	2
4	74.	How do you distinguish acetanilide and N-methyl benzamide by ^{13}C NMR?	2
4	75.	State Nitrogen Rule and mentions its importance in Mass spectroscopy.	2
3	76.	How do you differentiate Cis- and Trans- Protons in an olefin by Proton NMR?	2
4	77.	Integration of peaks is used in ^1H NMR but not in ^{13}C NMR. Why?	2
3	78.	N,N-Dimethylamine shows two signals for methyl groups in its ^{1}HNMRspectrum. Why ?	2
4	79.	How will you distinguish between 3-methyl and 4-methyl-cyclohexene with the help of mass spectrometry?	2
4	80.	Predict the masses of the ions produced in the mass spectrum of 2-heptanone by the McLafferty rearrangement.	2
4	81.	Ethyl benzoate in its mass spectrum show ions at M/Z = 88 and 60 and anabundant ion at M/Z = 71. Assign the structures for these fragments.	2
3	82.	Sketch the ^1H NMR spectra of pure and with acidic impurity of ethanol. Comment on their nature	2
4	83.	A compound with molecular formula C_4H_9Br gives the following signals in its^1H NMR spectrum. δ: 3.33 (d, 2H), 1.95 (m, 1H) and 1.04 (d, 6H). Deduce itsstructure.	2
3	84.	Write the multiplicity for the given organic compound in PMR 	2
4	85.	Show that toluene shows base peak at m/e=91.	2
4	86.	Write any two differences between ^{13}C and ^1H NMR.	2
4	87.	Identify the structure of organic compound with molecular formula $C_5H_{10}O$ by using PMR data δ: 0.95(6H, d, *J*=7hz), 2.10(3H, s), 2.43(1H,	2

		m, *J*=7Hz).	
4	88.	Show the fragmentation pattern of n-butylbenzene whose m/e values are 134, 92, 91(base peak), 65.	2
4	89.	In mass spectrum, highlight the importance of base peak.	2
4	90.	In ^{13}C NMR spectroscopy usually proton decoupled spectrum is taken. Give reason.	2
1	91.	Why there is no absorption due to n to σ^* transition in the spectrum of trimethyl amine in the acidic solution	4
1	92.	The absorption maximum for methyl chloride is 173 nm while for methyl iodide it is 259 nm. Why?	4
1	93.	Why hydrogen bonding shifts the absorption to shorter wavelengths in alcohols and amines?	4
1	94.	Amines absorb at higher wavelength than alcohols. Why?	4
1	95.	Name the electronic transition possible when UV light is absorbed	4
1	96.	What determines the wavelength of UV absorption by organic compound	4
1	97.	Prove that A = 2-log T	4
1	98.	Why do most photometers and spectrophotometers do not give linear read out in transmittance rather a linear read out in absorbance	4
1	99.	The molar absorptivity of a particular solute is 2.1×10^4. Calculate the transmittance through a cuvette with 5.0 cm light path for a 2.0 μM solution.	4
1	100.	What types of electronic transition are possible for each of the following compounds Dimethyl ether and methyl vinyl ether	4
1	101.	What types of electronic transition are possible for each of the following compound Triethyamine and cyclohexane	4
1	102.	What types of electronic transition are possible for each of the following compound Acetaldehyde and cyclopentene	4
1	103.	Calculate the absorbance and transmittance of a 0.002 M solution of a substance having molar absorption coefficient of 460 M^{-1} cm^{-1}. The	4

measurement is made by using a curvature of 1cm pathlength.

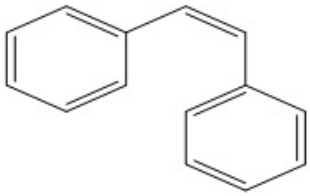
1	104.	The UV-Vis spectra in the solution phase have band structure whereas in the vapour phase we get sharp lines. Explain	4
1	105.	The intensity of light beam is reduced by 50% on passing through 1.0 cm of absorbing medium. What will be the reduction after going through 2.0 cm of the same medium?	4
1	106.	What would be the wavelength of absorption maximum of the certain compound CH_3CH_2=CHC(CH_3)=O using woodward-Fieser rule	4
1	107.	The intensity of light beam is reduced by 50% on passing through 1.0 cm of absorbing medium. What will be the reduction after going through 2.5 cm of the same medium.	4
1	108.	Predict the transition involved in a) methyl iodide b) methyl iodide c) methanol d) trimethylamine	4
1	109.	The position of absorption of acetone shifts in different solvents are: 279 nm in hexane, 272 nm in ethanol and 264.5 nm in water. Why?	4
1	110.	The C=C generally absorbs an absorption band at about 180-190 nm while C=O group in aldehydes and ketones show absorption maximum near 270-290 nm. Why?	4
1	111.	The bands due to due to n to π^* transition in amine disappear in acid solution. Why?	4
1	112.	Biphenyl shows the following ultraviolet absorption data. In its 2,2'-dimethyl derivative the absorption pattern becomes almost similar to *O*-xylene. Why?	4
1	113.	Why a polar solvent shifts n to π^* transition to shorter wavelength and π to π^* transition to longer wavelength?	4
1	114.	Explain why spectra of solutions in the UV-Vis regions tend to consist of a few broad peaks whereas the same solution measured in IR show many sharp peaks	4
1	115.	Explain why the UV spectrum of benzene vapour contains many sharp whereas the same spectrum of benzene solution in alcohol lacks fine detail and consists only broad peaks	4
1	116.	What is the absorbance of solution containing 3 ppm if transmittance was 65% in 10 mm cuvette	4

1 117. Calculate the molar concentration and absorptivity of a certain compound (10 ppm) which had an absorbance of 0.650 at 310 nm, while unknown compound (Molecular mass = 272.8) had on an absorbance of 0.930 4

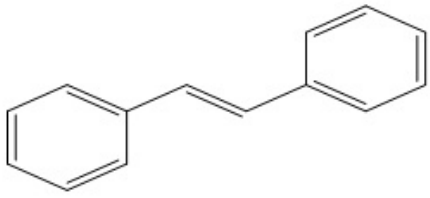
1 118. Calculate the molar absorptivity of complex of iron with 1,10-phenanthroline complex. If Fe (II) solution (4.7 ppm) when treated with reagent (made up to 100 ml) gave an absorbance of 0.467 at 510 nm 4

1 119. Calculate the concentration of the unknown solution of potassium dichromate if the transmittance measured was 75%. When one ml of 0.015 N potassium dichromate was added to 5 ml cell the reading changed to 65% T. 4

1 120. How will you distinguish between cis-stilbene and trans-stilbene by the use of UV-Vis spectroscopy



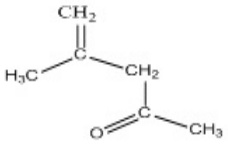
Cis stilbene



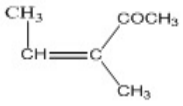
Trans stilbene

4

1 121. Mesityloxide, a condensation product of acetone is known to consist of two isomers (I) and (II) shown below. One isomer exhibits a maximum at 235 nm with ε= 12000 while other shows no high intensity absorption above 220 nm. Which of the two isomers absorbs at 235 nm? and Why?



I



II

4

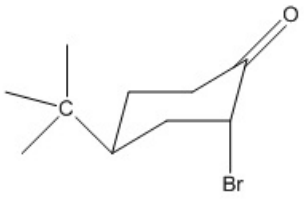
1 122. The UV Spectrum of acetone shows absorption bands at 166, 189 and 279 nm. What type of transition is responsible for each of these bands? 4

1 123. Chloromethane has an absorption maximum at 172 nm, bromomethane shows an absorption at 204 nm and iodomethane shows band at 258 nm. What type of transition is responsible for each band? How can the trend of absorption can be explained? 4

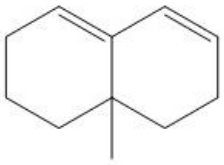
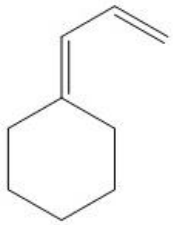
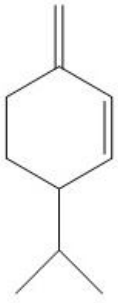
1	124.	Explain different types of electronic transitions involved in CH_3CONH_2 with the help of molecular orbital energy level diagram	4
1	125.	The molar absorptivity of a substance is 1.5 x 10^4 cm^{-1} mol^{-1} dm^3. Calculate the transmittance through a cuvette of path length 50 mm containing 2 x 10^{-6}mol dm^{-3} solution of the substance.	4
1	126.	Explain the effect of solvents on the π-π^* and n-π^* transitions	4
1	127.	Discuss the bathochromic shift of conjugated dienes based on HOMO-LUMO theory	4
1	128.	Explain the how UV-Vis spectra can be used to differentiate between ethylene and butadiene by HOMO-LUMO theory	4
1	129.	Discuss briefly different types of absorption bands in UV-Vis spectra.	4
1	130.	What is the magnitude of molar absorptivity if a certain compound had a molecular mass = 92 if 0.015% solution by weight had 27% T with 10 mm cells?	4
1	131.	What should be the magnitude of molar absorptivity if a solution was prepared by dissolving 0.1298 g of compound (molecular mass = 214.3) in one liter of solution. The absorbance of the new solution at 255 nm was 0.824?	4
1	132.	A solution containing 8.75 ppm $KMnO_4$ has a transmittance of 0.743 in a 1.00 cm cell at 520 nm. Calculate the molar absorptivity of $KMnO_4$	4
1	133.	Beryllium(II) (atomic mass : 9.01) forms a complex with acetylacetone (166.2 g/mol). Calculate the molar absorptivity of the complex, given that a 1.34 ppm solution has a transmittance of 55.7% when measured in a 1.00 cm cell at 295 nm, the wavelength of maximum absorption.	4
1	134.	A 5.00 ml aliquot of a solution contains 5.94 ppm iron(III) is treated with an appropriate excess of KSCN and is diluted to 50.0 ml. what is the absorbance of the resulting solution at 580 nm in a 2.50 cm cell (ε=7.00 $\times 10^3$ L mol^{-1} cm^{-1})	4
1	135.	When 5 mg of a certain amine picrate is dissolved in 100 ml of ethanol, its absorbance is measured in a 2.00 cm cell at 380 nm as 0.415. What is the molecular weight of the amine picrate? Given: ε= 13400 M^{-1} cm^{-1}	4
1	136.	Calculate the concentration in μg ml^{-1} of a solution of organic compound (mol mass 211.0) in 0.11 M HCl giving an absorption at its λ_{max} 281 nm of 0.612 in 4 cm cell. The molar absorptivity at 281 nm is 5372.	4

1	137.	In the determination of acetone in biological fluids, water blank instead of a reagent blank used. A 2.5 mg% standard gave an absorbance of 0.81 and unknown gave 0.6. In a separate determination it was found that the reagent blank gave an absorbance of 0.09 when read against the water blank. Calculate the concentration of the acetone in the unknown.	4
1	138.	The absorption spectrum for titanium peroxide complex ion in perchloric acid showed a maximum of 400 nm. The absorbance of a 32.0 μg/ml of titanium gave an absorbance of 0.456. An unknown solution treated in an identical fashion gave an absorbance of 0.501. assuming identical cells, find out the concentration of the unknown.	4
1	139.	Draw the UV-Vis energy level transitions for acetaldehyde, formaldehyde and acrylaldehyde.	4
1	140.	The wave number of absorption due to carbonyl increases in acid halides. Discuss the change in the absorption with a suitable explanation	4
2	141.	How an acid unhydride can can be detected by IR spectroscopy.	4
2	142.	Distinguish between maleic acid and fumaric acid by IR spectroscopy	4
2	143.	Which of the following molecules will show IR spectrum HCl, CH_4, CO_2, H_2 and N_2O?	4
2	144.	Give the absorption of the following groups in the IR region, OH in phenol; CO in aldehydes; CN in aromatic nitriles; OH stretching in carboxylic acid	4
2	145.	Why methanol is a good solvent for UV but not for IR determination?	4
2	146.	What is the necessary condition for a molecule to absorb IR radiation	4
2	147.	Why C=O stretching for salicylic acid occurs at 1665 cm^{-1} while p-hydroxybenzoic acid it occurs at 1680 cm^{-1} ?	4
2	148.	Why translational motion is not involved in molecular spectra?	4
2	149.	Why absorption of ultraviolet and visible mechanism can be studied together but IR absorption studies are made separately?	4
2	150.	Estimate the frequency of absorption of the cyanide stretch in HCN	4
2	151.	Calculate the wavelength of absorption of HCl stretching frequency	4
2	152.	Explain the effect of different groups on the wavenumber of C=O	4

		absorption in the case p-quinones	
2	153.	Explain the frequency of OH stretching is observed at higher range than the C-H stretching	4
2	154.	When will the Raman spectrum result?	4
2	155.	Using 4358 A^o lines of mercury as the sources of radiation a Raman line was observed at 4447 A^o. What is the Raman shift?	4
2	156.	State the factors that complicate the IR spectrum.	4
2	157.	Calculate the wave number of a band and energy of separation at two levels if transition between two vibrational energy levels of a molecule gave an IR peak around 6.43 mm.	4
2	158.	Find out the wavelength (nm) and wave number (cm^{-1}) of IR for below frequencies, 1.096×10^{14} Hz.	4
2	159.	The IR spectrum of CO shows a vibrational absorption peak at 2170 cm^{-1} a) what is the force constant for the CO bond? b) What wave number would the corresponding peak for ^{14}CO occur?	4
2	160.	The force constant for the band in HF is about 9×10^{15}/cm. a) calculate the vibrational absorption peak for HF b) calculate the absorption peak for DF	4
2	161.	Carbon monoxide shows an absorption band at 2140 cm^{-1}. Calculate the force constant for this molecule (C=12, O=16). Express this result in Nm^{-1}.	4
2	162.	Calculate the wave number of the C-Cl band in carbon tetrachloride. The force constant is 3.4×10^5dyn/cm	4
2	163.	The IR absorption spectrum of HCN shows three absorption bands at 3312, 2089, and 712 cm^{-1}. Deduce from this whether HCN is linear or bent. Assign the absorption band values	4
2	164.	In case of α-halogenocyclohexanone, there is a rise of C=O absorption by 20 cm^{-1} when the halogen atom is equatorial whereas no such shift is observed when the halogen atom is in the axial position. Explain	4
2	165.	Why enolic form of β-diketones exhibit much lower C=O absorption than that for conjugated ketones?	4
2	166.	An organic compound with molecular formula C_3H_7NO shows absorption data in the region 3415, 3236, 3030-2899, 1668, 1635, and 1460 cm^{-1}. Give the probable structure of the molecule	4

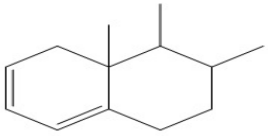
2	167.	The IR spectrum of methyl salicylate shows peak at 3300, 1700, 3050, 2990, 1590 and 1540 cm^{-1}. Attribute these peaks to the following structures i) Methyl group ii) C=O iii) O-H group in the ring iv) aromatic ring	4
2	168.	Compare the actual resonance vibration group frequency (in Hz) for diatomic molecules if force constant (10×10^5dyn cm^{-1}) for C=N and C=O. Explain your answer with a suitable explanation for shift in the wavenumber of the bonding systems.	4
2	169.	Distinguish between the following each pair using IR spectra a) Aniline and N-methylaniline b) *o*-hydroxybenzoic acid and *m*-hydroxybenzoic acid	4
2	170.	Which form of 2-bromo-4-tertiarybutylcyclohexanone shows higher C=O absorption and why?  cis form  Trans form	4
4	171.	Describe DEPT. Discuss its applications.	4
4	172.	Describe Off-resonance decoupled, Proton decoupled ^{13}C NMR experiments.	4
3	173.	Sketch the ^{1}H NMR spectra of pure and with acidic impurity of ethanol. Comment on their nature.	4
4	174.	Discuss the basic principle of Mass Spectroscopy.	4
4	175.	Sketch Off-resonance decoupled and Proton decoupled ^{13}C spectra of 4-methyl acetophenone	4
4	176.	Illustrate the genesis and applications of isotope ion peaks in interpretation of MS Spectra.	4
3	177.	Give an account on vicinal coupling and germinal coupling.	4
4	178.	Write briefly about FAB method of ionization and give its advantages over other methods of ionization.	4
4	179.	Write briefly about Field Desorption method of ionization and give its	4

		advantages over other methods of ionization.	
4	180.	Explain the mechanism of McLAfferty rearrangement taking suitable example.	4
4	181.	Write briefly about Off resonance decoupled and Complete proton decoupled ^{13}C NMR spectrum.	4
3	182.	Account on Nuclear Overhauser Effect	4
3	183.	Write a note on Shift Reagents.	4
3	184.	Discuss Anisotropic effects prevalent in ^{1}H NMR for Alkenes and Alkynes.	4
4	185.	Discuss McLAfferty rearrangements in the mass spectral study of aldehydes and ketones.	4
4	186.	Write a note on ^{19}F NMR	4
4	187.	Elaborate on principle and application of ^{31}P NMR.	4
4	188.	Explain the principle and application of ^{15}N NMR.	4
4	189.	Discuss McLAfferty rearrangement with examples.	4
4	190.	How will you differentiate among o-, m- and p-xylenes on the basis of their proton decoupled ^{13} C NMR?	4
4	191.	Deduce the structure of the compound on the basis of the following data and assign the values. Molecular formula $C_7H_6O_2$ IR 3239(br), 2849, 2756(vw), 1665, 1456, 1281, 1156, 760 cm^{-1} ^{1}H NMR δ 7.01 (m, 2H), 7.8 (m, 2H) and 9.89 (s, 1H) and 10.95 (s, 1H) exchanges with D_2O.	4
4	192.	What do you understand by metastable ions or peaks? How these are recognized in the mass spectrum andwhat is their importance?	4
4	193.	Explain McLAfferty rearrangement in following compounds- i. $(CH_3CH_2CH_2)_2CO$ ii. $CH_3CH_2CH_2COOCH_3$	4
3	194.	Briefly explain instrumentation of nuclear magnetic spectrometer.	4
4	195.	Deduce the structure of organic compound with molecular weight 150 shows the following spectral data: UV: λ_{max} 276 nm; IR: 3030-2979, 1695, 1692 cm^{-1} ^{1}H NMR: 2.1(s, 3H), 3.85(s, 3H), 7.2(d, 2H) and 7.65(d, 2H)	4

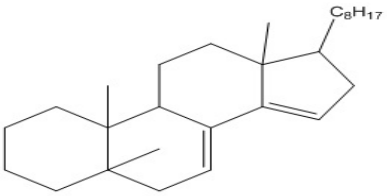
3	196.	Write a note on FT NMR.	4
3	197.	Explain how simplification of complex spectra can be achieved by isotopic substitution and by increasing magnetic field strength.	4
4	198.	Deduce the structure of the compound ($C_9H_{10}O$) assigning all the spectral data given: UV: λ_{max} 260 nm; IR: 1715 cm^{-1} ^1H NMR δ: 2.1(s, 3H), 3.7(s, 2H) 7.1-7.5 ppm (5H, m).	4
4	199.	An organic compound with molecular formula C_4H_8O showed following spectral data; UV: λ_{max} 274 nm ε= 17; IR: 1715 cm^{-1}(s); ^1H NMR δ: 1.07(s, 18 sq), 2.12(q, 17.6 sq) 2.48 ppm (s, 12 sq). Deduce the structure.	4
4	200.	Assign the structure of the compound ($C_7H_{14}O_2$) whose spectral data are given UV: λ_{max} 278 nm (ε= 30); IR: 2990, 2920, 2830, 1705, 1380, 1220, and 1090 cm^{-1} ^1H NMR δ: 2.1(s, 3H), 3.7(q, 2H) 7.1-7.5 ppm (5H, m).	4
1	201.	Predict on the basis of the HMO theory, the wavelength of the π to π^* transition observed in the UV spectra of the conjugated linear polyenes, butadiene, hexatriene and octatetraene	6
1	202.	Discuss the general application of UV-Vis spectroscopy in organic chemistry.	6
1	203.	Discuss the any three application of UV-Vis spectroscopy	6
1	204.	α-cyperone exists as Using woodward rules decide which among the above structures is consistent with the observed value of 252 nm	6
1	205.	Following the woodwardfieser rules, calculate the absorption maximum for each of the following compounds a.  c.  b.  d. 	6
1	206.	Using Woodward's rules, predict the wavelength of maximum absorption	6

for each of the following compounds

a.

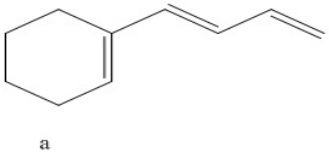


b.

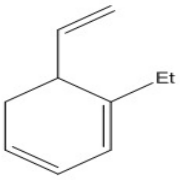


1 Explain how ultraviolet spectrum can be used to decide between the following isomeric systems

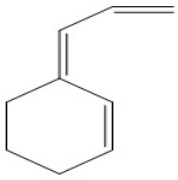
207. 6



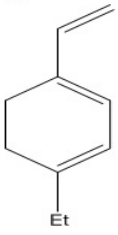
a



c



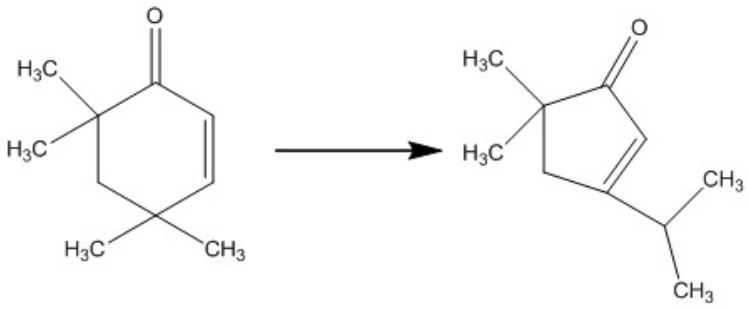
b



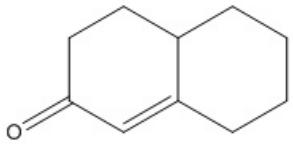
d

1 Predict the UV maximum for the reactant and product of the following photochemical reaction:

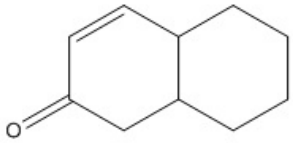
208. 6



1 209. Predict and explain whether UV Visible spectroscopy can be used to distinguish between the following pairs of compounds. If possible support your answer with calculations 6

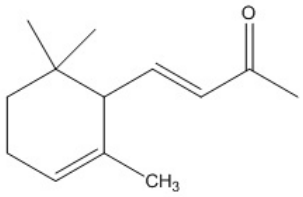


a

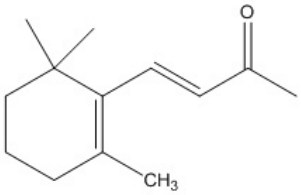


b

1 Predict and explain whether UV Visible spectroscopy can be used to distinguish between the following pairs of compounds. If possible support your answer with calculations

210. 

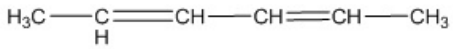
a

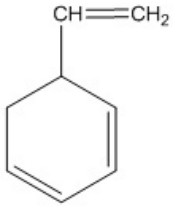


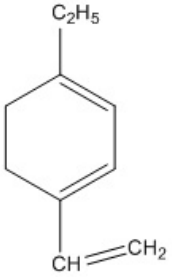
b

6

1 Calculate the absorption maximum for the following compounds by woodward-Fieser rules

211. a. 

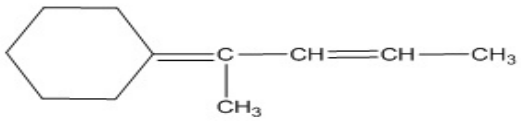
b. 

c. 

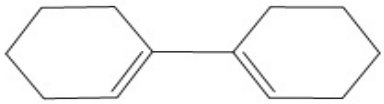
6

1 212. Calculate the absorption maximum for the following compounds by woodward-Fieser rules 6

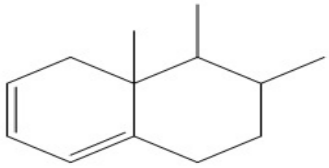
a.



b.



c.

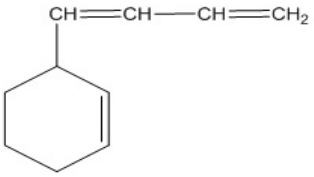


1

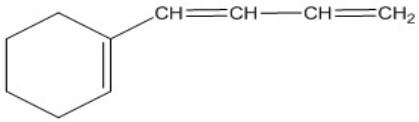
Calculate the absorption maximum for the following compounds by woodward-Fieser rules

213.

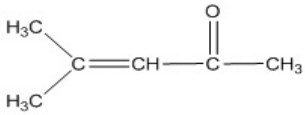
a.



b.



c.



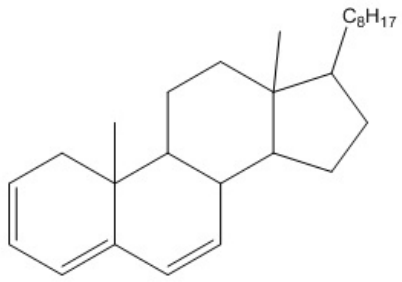
6

1

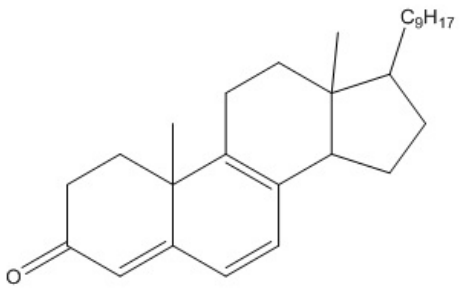
214. Calculate the absorption maximum for the following compounds by woodward-Fieser rules

6

a.

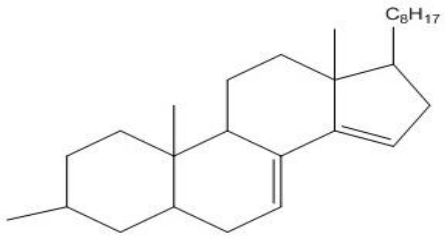


b.



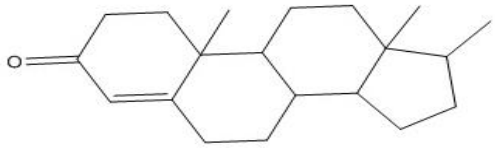
1 Calculate the absorption maximum for the following compounds by woodward-Fieser rules

215. a.



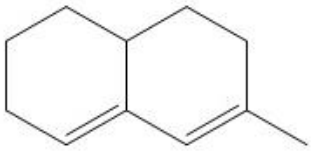
6

b.

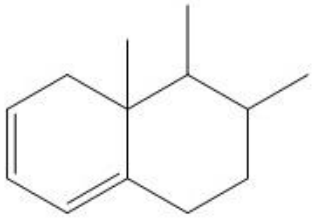


1 216. Calculate the absorption maximum for the following compounds by woodward-Fieser rules 6

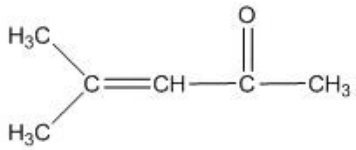
a.



b.

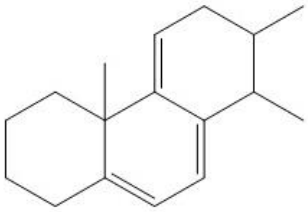


c.



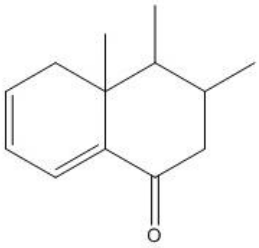
1 Calculate the absorption maximum for the following compounds by woodward-Fieser rules

a.



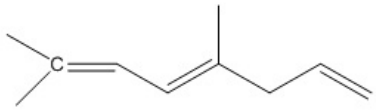
217. 6

b.

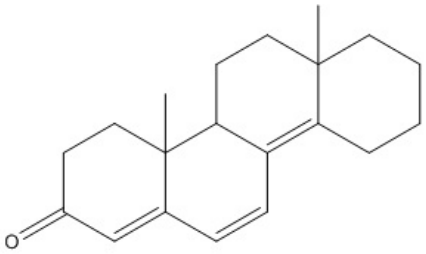


1 218. Calculate the absorption maximum for the following compounds by woodward-Fieser rules 6

a.



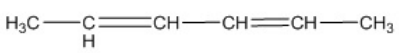
b.



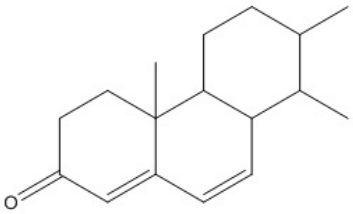
1 Calculate the absorption maximum for the following compounds by woodward-Fieser rules

219. 6

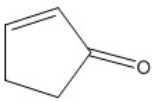
a.



b.

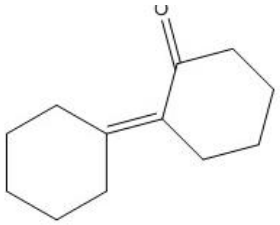


c.

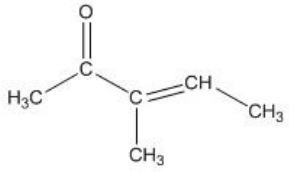


1 220. Calculate the absorption maximum for the following compounds by woodward-Fieser rules 6

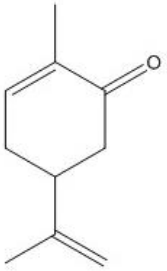
a.



b.



c.



1

221. At 580 nm, the wavelength of its maximum absorption, the complex $FeSC^{N2+}$ has a molar absorptivity of 7.00×10^3 L mo^{l-1} cm^{-1}, calculate

a) The absorbance of a 3.75×10^{-5} solution of the complex at 580 nm in a 1.00 cm cell

b) The absorbance of a solution in which the concentration of the complex is twice that in a)

c) The transmittance of the solutions described in a) and b)

d) The absorbance of a solution that has half the transmittance of that described in (a)

6

1

222. A solution containing the complex formed between Bi(III) and thiourea has a molar absorptivity of 9.32×10^3 L mo^{l-1} cm^{-1} at 470 nm

a) What is the absorbance of a 6.24×10^{-5} M solution of the complex at 470 nm in a 1.00 cm cell?

b) What is the percent transmittance of the solution described in (a)?

c) What is the molar concentration of the complex in a solution that has the absorbance described in (a) when measured at 470 nm in a 5.00 cm cell?

6

1

223. The complex formed between Cu(I) and 1,10-phenanthroline has a molar absorptivity of 7000 L $cm^{-1}mol^{-1}$ at 435 nm, the wavelength of maximum absorption. Calculate

a) The absorbance of a 6.77×10^{-5} M solution of a complex when

6

measured in a 1.00 cm cell at 435 nm

b) The percent transmittance of a solution in (a)

c) The concentration of a solution that in a 5.00 cm cell has the same absorbance as the solution in (a)

d) The path length through a 3.40×10^{-5} M solution of the complex that is needed for an absorbance that is the same as the solution in (a)

1 224. A compound X is to be determined by UV/visible spectrophotometry. A calibration curve is constructed from the solution of X with the following results: 0.50 ppm, A=0.24; 1.5 ppm, A=0.36; 2.5 ppm, A=0.44;3.5 ppm, A=0.59; 4.5 ppm, A=0.70. Find the slope and intercept of the calibration curve. Construct a plot of the calibration curve and determine the unknown concentration by hand from the plot. 6

1 225. One common way to determine phosphorus in urine is to treat the sample, after removing the protein, with molybdenum (VI) and then reduce the resulting 12-molybdophosphate complex with ascorbic acid to give an intense blue coloured species called molybdenum blue. The absorbance of the molybdenum can be measured at 650 nm. A patient produced 1122 ml of urine in 24 hrs. A 1.00 ml aliquot of the sample was treated with Mo(IV) and ascorbic acid and was diluted to a volume of 50.00 ml. The calibration curve was prepared by treating aliquots of phosphate standard solutions in the same manner as the urine sample. The absorbances of the standards and the urine sample were obtained at 650 nm, and the following results were obtained. 6

Solution	**A_{650}**
1.00 ppm P	0.230
2.00 ppm P	0.436
3.00 ppm P	0.638
4.00 ppm P	0.848
Urine sample	0.518

a) Find the slope, intercept. Construct a plot of calibration curve. Determine the number of ppm of P in the urine sample. Compare the unknown concentration with that obtained manually from the calibration curve

b) What mass in grams of phosphorus was eliminated per day by the patient?

c) What is the phosphate concentration in urine in mM?

1 226. Nitrite is commonly determined by a colorimetric procedure using a reaction called the Griess reaction. In this reaction, the sample containing nitrite is reacted with sulphanilide and N-(1-naphthyl)ethylenediamine to form a colored species that absorbs at 550 nm. By using an automated flow 6

analysis instrument, the following results were obtained for standard solutions of nitrite and for a sample containing an unknown amount.Construct a plot of the calibration curve

Solution (μM)	A_{550}
2.0	0.065
6.0	0.205
10.0	0.338
14.0	0.474
18.0	0.598
unknown	0.402

a) Find the slope, intercept of the calibration curve

b) Determine the concentration of nitrite in the sample

1 In the determination of acetone in biological fluids, the following calibration curve between absorbance and concentration was obtained.

Acetone standards (mg/100 ml)	*A*
Reagent blank	
0.5	
1.0	
2.0	
4.0	
6.0	
8.0	

6

227. Calculate the molar absorptivities and specific absorptivities

2 The bent triatomic molecule O-N-Cl has three strong IR bands at 1799, 592, 332 cm^{-1}. Diagram the fundamental modes of vibration of O-N-Cl and assign absorption bands to each of the wave numbers 6

228.

2 Diagram the fundamental modes of vibration of carbondioxide and predict which modes will be IR active and which will be Raman-active 6

229.

2 Diagram all the modes of vibration of planar N_2O_4 and predict whether all the modes the vibration are Raman or IR active 6

230.

2 A triatomic molecule XY_2 has three fundamental vibrational frequencies; two of them are IR active and third is Raman active. What conclusion can be drawn regarding the structure of the molecule 6

231.

2 Diagram all the modes of vibration of ethylene and predict which of the modes will be IR active and which will be Raman active. Indicate your answer with suitable explanation for change in the dipole moment and 6

232.

		polarizability	
2	233.	Diagram all the modes of vibration of cyanogens $(CN)_2$ and predict which modes IR active and which will be Raman active. Indicate your answer with suitable explanation for change in the dipole moment and polarizability	6
2	234.	Explain inductive effect, hydrogen bonding and electrical effects on IR bands.	6
2	235.	Discuss any three factors affecting IR spectra of organic molecules.	6
2	236.	Explain the wave theory of Raman spectroscopy.	6
2	237.	Discuss the quantum theory with reference to Raman spectroscopy	6
2	238.	The IR spectrum of nitrogen di oxide reveals that three strong bands at 1616 cm^{-1}, 1323 cm^{-1}, 750 cm^{-1}. On the basis of this spectrum what	6
2	239.	Sketch all the vibrations of carbon di oxide and water molecules and discuss its IR and Raman activity of vibrations	6
2	240.	Explain classical and quantum theory of Raman spectroscopy.	6
2	241.	How will you differentiate between the following compounds by IR spectroscopy. Provide the IR data for differentiation • Benzaldehyde and acetaldehyde • Benzene and cyclohexane • Acetic acid and benzoic acid	6
2	242.	How will you differentiate between the following compounds by IR spectroscopy? Provide the IR data for differentiation • Acetyl amine and acetonitrile • Acetone and acetic acid • Ethanoic acid and ethanol	6
2	243.	How will you differentiate between the following compounds by IR spectroscopy? Provide the IR data for differentiation • Benzoic acid and acetophenone • Formaldehyde and acetaldehyde • Formic acid and Formaldehyde	6
2	244.	Distinguish between the following each pair using IR spectra • Acetone and acytylene • Ethanal and ethanol • Acetonitrile and acetamide	6
2	245.	How will you differentiate between the following compounds by IR spectroscopy? Provide the IR data for differentiation	6

- benzylamine and N,N-dimethylacetamide
- Propanal and 2-propenal
- *o*-hydroxybenzoic acid and *p*-hydroxybenzoic acid

2 Indicate the whether the following vibrations will be active in the IR spectroscopy. Indicate the vibration modes and justify the answer with proper explanation

246. 6

Molecular	**Modes of vibration**
Ethane	C-C stretching
1,1,1-tricholoroethane	C-C stretching
Sulphurdioxde	Symmetric stretching
Water	Symmetric stretching
Carbondioxide	Unsymmetrical stret

2 Account for the following variation in IR bands of carbonyl stretching frequency

247. 6

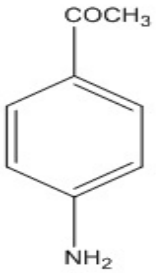
Group	**C=O absorption Wave number (cm^{-1})**
Acids	1650
Amides	1670
Ketones	1700
Esters	1735
Acid chlorides	1800

2 Account for the following variation in IR bands

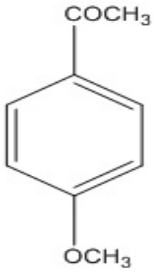
248. 6

Hydrocarbon	Absorption frequencies, cm^{-1}	
	C=C stretch	**C-H Stretch**
Cyclohexene	1646	3017
Cyclopentene	1611	3045
Cyclobutene	1571	3070

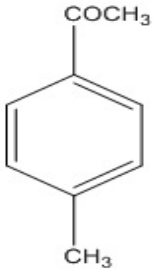
2 249. State the C=O absorption in the following compounds with suitable explanation (values are provided in terms of cm^{-1}) 6



1677

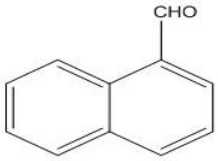


1684

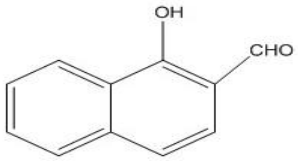


1693

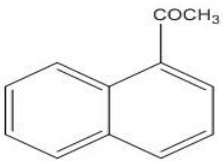
2 Explain the lowering of C=O absorption in the following compounds (values in the brackets are provided in terms of cm^{-1})

250. 

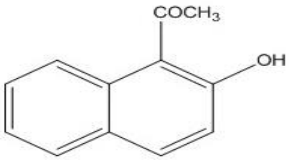
1700



1651



1685



1625

6

2 In case of following molecules, the vibration will be active or inactive in the IR region. Discuss the each of the following vibration with suitable explanation

251.

Sulphurdioxide	Symmetric stretcl
Ethane	C-C stretching
Ethene	C-H stretching
Ethyne	C-C stretching

6

4 252. How will you distinguish n-butanol, sec-butanol, tert-butanol and iso-butanolusing ^{13}C NMR spectroscopy. 6

3 253. Explain the factors which influence chemical shifts in NMR Spectroscopy. 6

3 254. Give Reason: a. Why aromatic protons appear at higher δ value? b. Acetylenic protons absorb at higher field than olefin protons 6

4 255. Write a note on NOESY. 6

4	256.	Deduce the structure of organic compound with molecular formula $C_9H_{10}O_2$ based on following spectral data IR= 1690 cm^{-1} PMR δ=2.5 (s, 3H) , 3.98 (s, 3H), 7.08 (d, 2H, J=12 Hz), 8.18 (d, 2H, J=12 Hz) ppm 13C = 27, 55, 114, 125, 130, 165, 197 ppm	6
4	257.	Deduce the structure of the compounds on the basis of the following data and assign the values. Molecular Formula: $C_7H_6O_2$, UV λ_{max} (EtOH) 258 nm (E= 20,000), IR 3239 (br), 1665,1459, 1281, 1156, 760 cm^{-1},^1H NMR δ= 7.01 (m, 2H), 7.8 (m, 2H), and 9.89 (s, 1H) and 10.95 (s, 1H) exchanges with D_2O; Mass m/e (relative abundance): 122 (74), 121(100), 104(15), 93(22), 76(12), 65(26) and 63(08).	6
4	258.	Explain different ionization techniques used in mass spectroscopy.	6
4	259.	Discuss mass fragmentation pattern for Alcohols.	6
3	260.	Explain the factors which influence chemical shifts in NMR Spectroscopy.	6
3	261.	Give Reason: a. Why aromatic protons appear at higher δ value? b. Acetylenic protons absorb at higher field than olefin protons	6
4	262.	Write a note on COSY.	6
4	263.	Deduce the structure of organic compound with molecular formula $C_9H_{10}O_2$ based on following spectral data IR= 1690 cm^{-1} PMR δ=2.5 (s, 3H) , 3.98 (s, 3H), 7.08 (d, 2H, J=12 Hz), 8.18 (d, 2H, J=12 Hz) ppm 13C = 27, 55, 114, 125, 130, 165, 197 ppm	6
4	264.	Deduce the structure of the compounds on the basis of the following data and assign the values. Molecular Formula: $C_7H_6O_2$, UV λ_{max} (EtOH) 258 nm (ε= 20,000), IR 3239 (br), 1665,1459, 1281, 1156, 760 cm^{-1},^1H NMR δ= 7.01 (m, 2H), 7.8 (m, 2H), and 9.89 (s, 1H) and 10.95 (s, 1H) exchanges with D_2O; Mass m/e (relative abundance): 122 (74), 121(100), 104(15), 93(22), 76(12), 65(26) and 63(08).	6
4	265.	Explain different ionization techniques used in mass spectroscopy.	6
3	266.	Explain the instrumentation of Nuclear magnetic resonance spectrometer using diagram and discuss about sampling.	6
4	267.	Discuss fragmentation pattern of Hydrocarbons.	6
4	268.	Discuss mass fragmentation pattern observed in amines.	6
4	269.	Deliberate on mass fragmentation pattern of aldehydes and ketones.	6

4	270.	Elaborate on Ortho effect and Hydrogen transfer rearrangements using examples.	6
4	271.	Deliberate on mass fragmentation of acids and esters.	6
4		Assign the structure of compounds using given data below	
	272.	Molecular Formula $C_7H_{14}O_2$; UV λ_{max}278(30) nm; IR: 2990, 2920, 2830, 1705, 1380, 1220, and 1090 cm^{-1}. ^1H NMR δ= 3.2 (s, 3H), 2.5 (s, 2H), and 2.2 (s, 3H) and 1.2 (s, 6H)ppm; ^{13} C (off resonance decoupled) 26(q), 32(q), 50(t), 54(q), 74(s), 204(s); MS m/e= 130(2), 115(18), 100(10), 73(85), 43(100).	6
4		Assign the structure of compounds using given data below	
	273.	a. Molecular Formula C_6H_5Cl; IR: 3100, 1490, 700-800 cm^{-1}. ^1H NMR δ= 7.1-7.4 (m, 4H) ppm; ^{13} C 123,129, 130, 135, 135 ppm; MS m/e= 114, 112 & 77. b. Molecular Formula $C_{10}H_{14}$; IR: 3000, 1500, 900 cm^{-1}. ^1H NMR δ= 7.4 (s, 2H), 2.2 (s, 3H), 2.1(s, 3H), 1.9 (s 6H) ppm; ^{13} C 15, 20, 21, 129, 132, 136, 138 ppm; MS m/e= 134, 119, 89 & 74.	6
4	274.	Explain the working principle of time of flight and Quadrapole mass separators	6
4	275.	Elaborate on various experiments employed in ^{13}C NMR Spectroscopy.	6
4	276.	Discuss EI and CI ionization techniques used in mass spectroscopy.	6

Question Paper Pattern- Model Question Paper

Q.P Code:14MSCCH130

St. Philomena's College (Autonomous) Mysore
III Semester M.Sc. – C3 - Final Examination October 2017
Subject: CHEMISTRY
Title: ORGANIC SPECTROSCOPY (HC)

Time: 3 Hours **Max Marks: 70**

PART –A

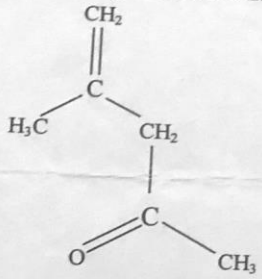
Answer the following Questions: **10x2=20**

1. a. Convert 600 nm into eV. Given that h = 6.627×10^{-34} Js, C = 3×10^{8} ms^{-1} and 1 eV = 1.602×10^{-19} J.

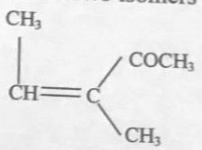
   b. Define mutual exclusion principle with an example.

   c. What are the different types of absorption bands in UV-Vis spectra.

   d. Mesityloxide, a condensation product of acetone is known to consist of two isomers (I) and (II) shown below. One isomer exhibits a maximum at 235 nm with $\varepsilon = 12000$ while other shows no high intensity absorption above 220 nm. Which of the two isomers absorbs at 235 nm? and Why?



I



II

   e. What is the condition for NMR active nucleus?

   f. Acetic acid and Phenol will give how many proton NMR signals?

   g. Write the proton NMR spin multiplicities for *n* – Hexanal.

   h. What is Nitrogen rule? Explain.

   i. A peak appears at 150 Hz away from TMS in 100 MHz spectrometer. Find the chemical shift values in ppm.

   j. Why TMS is used as reference standard?

PART –B

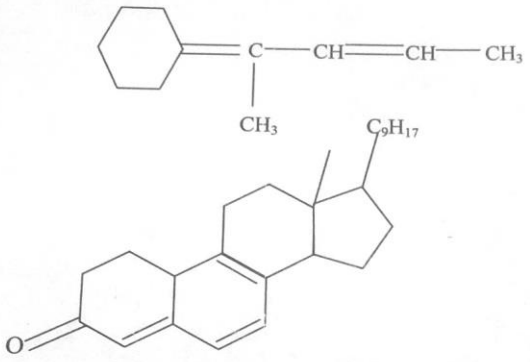
Answer FIVE of the following Questions: **5x10=50**

a. The IR spectrum of nitrogen di oxide reveals three strong bands at 1616 cm^{-1}, 1323 cm^{-1} and 750 cm^{-1}. On the basis of this spectrum what possible structure can be assigned to nitrogen di oxide? Mention the vibrations that can be assigned to these bands. **04**

b. Explain 19F NMR taking suitable examples. **06**

PTO

3. a. Calculate the λ_{max} for following compounds: **06**



b. Account for the following variation in IR bands: **04**

Hydrocarbon	Absorption Frequencies, cm^{-1}	
	C = C Stretch	C – H Stretch
Cyclohexene	1646	3017
Cyclopentene	1611	3045
Cyclobutene	1571	3070

4. a. What are the factors which affects chemical shift values? Discuss briefly. **05**

b. Discuss the types of relaxation process. **05**

5. a. An organic compound having molecular formula C_8H_8O showed the following spectral data *UV*: $\lambda_{max} = 274\,nm$; *IR* : $1715\,cm^{-1}$ (*strong*); *PMR* : $\delta = 1.07, t, 18$ squares. $\delta = 2.12, s, 17.6$ squares $\delta = 2.48, q, 12.0$ squares. Deduce the structure of the compound. **07**

b. Write a note on MRI. **03**

a. Elucidate the structure of the compound given below that meets the spectral data: **2.5x4=10**

a) C_2H_6O, 1 singlet b) $C_4H_8Cl_2O$, 2 triplet c) C_3H_7Cl, 1 doublet, 1 septet

d) $C_{10}H_{14}$, $\delta = 1.3, s, 9H.$, $\delta = 7.3, s, 5H$

a. Write a short note on instrumentation of Mass spectroscopy with schematic diagram. **05**

b. Explain Mc Lafferty Rearrangement with example. **05**

a. Explain the wave theory of Raman spectroscopy. **06**

b. Write the differences between H^1 NMR and C^{13} NMR. **04**

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